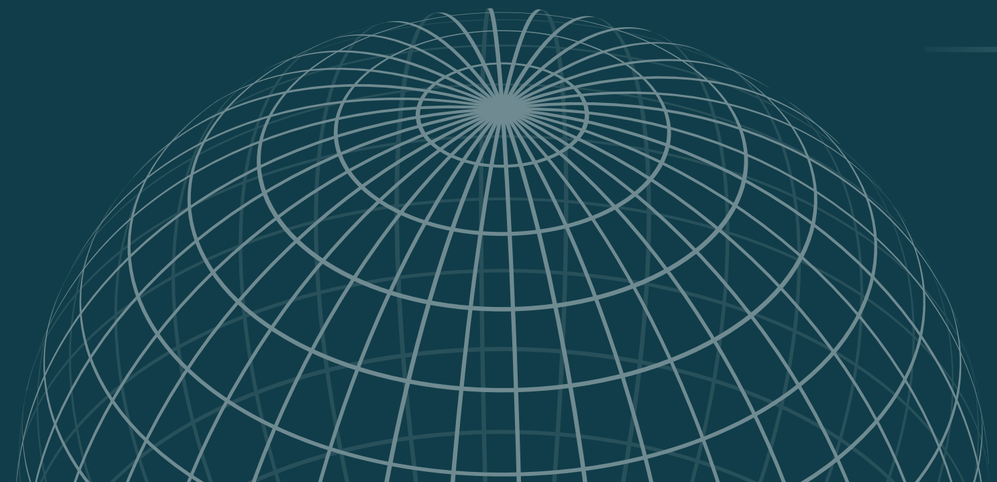
3MTT Deeptech Mentorship Program

Project Presentation

MATERNAI

Presented by Eniola Adetunji

MaternAI is an AI-powered maternal health risk prediction system that uses machine learning to classify maternal health risk based on selected clinical measurements.





Problem Statement

During pregnancy, several clinical measurements such as blood pressure, blood sugar, body temperature, and heart rate can provide important information about maternal health.

However, interpreting multiple clinical measurements together can be complex. Important patterns may not always be immediately obvious when each measurement is considered separately.

This creates a need for better ways to analyze clinical data and identify patterns associated with different levels of maternal health risk.

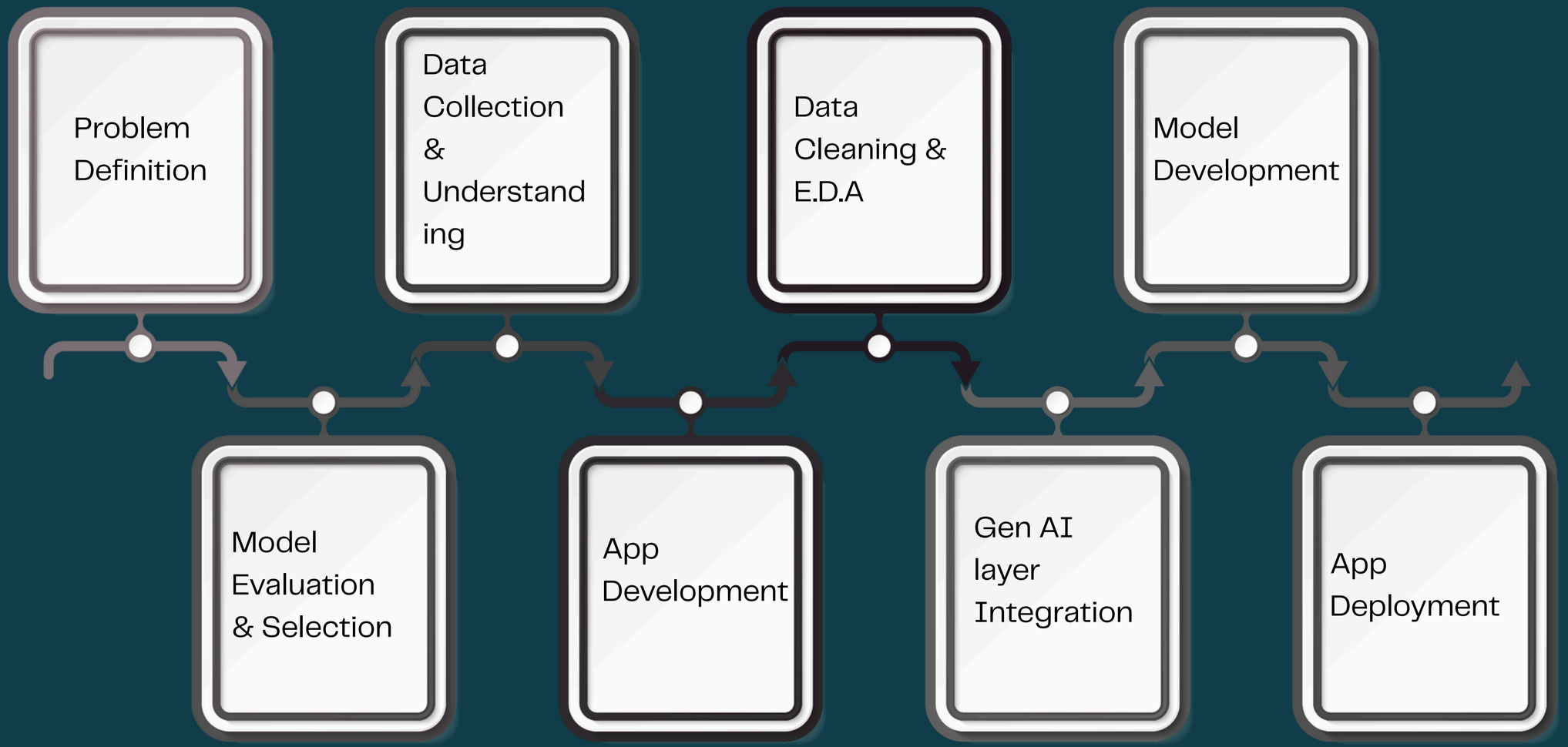
Project Aim & Objectives

This project aims to develop an AI-powered application that predicts maternal health risk from selected clinical measurements and provides an understandable explanation of the prediction.

Objectives:

1. Prepare maternal health data for machine learning.
2. Develop a machine learning model to classify maternal health risk.
3. Build an interactive web application for users to input clinical measurements and receive predictions.
4. Integrate generative AI to explain predictions in simple, understandable language.
5. Deploy the application as an accessible web-based tool.

Project workflow



The Data behind MaternAI

Link to Dataset

6 Clinical Features

Age

Systolic Blood Pressure

Diastolic Blood Pressure

Blood Sugar

Body Temperature

Heart Rate

1 target column

Risk Level

- 🟢 Low Risk
- 🟡 Mid Risk
- 🔴 High Risk

- 1014 rows
- 7 columns

Data cleaning

Link to Notebook

Missing values
No null values

Duplicates
562 identified → retained

Outliers
Potential outliers → retained

Data Types
Appropriate formats

The dataset required no major cleaning operations and was considered suitable for further analysis and model development.

Exploratory Data Analysis

Key Findings Include:

- Blood Sugar showed the clearest separation between maternal health risk categories, with high-risk pregnancies generally recording higher values.
- Systolic and Diastolic Blood Pressure also showed noticeable differences across the risk groups, with higher values generally associated with higher risk.
- Age showed some variation, with high-risk pregnancies tending to involve slightly older mothers, but substantial overlap existed between groups.
- Body Temperature and Heart Rate showed considerable overlap across the risk categories, suggesting weaker separation when considered individually.

Model Evaluation & Selection

Four classification models were developed and evaluated.

	Model	Accuracy	Precision	Recall	F1 Score
2	XGBoost	0.857143	0.865814	0.857143	0.858420
0	Decision Tree	0.837438	0.855618	0.837438	0.840048
1	Gradient Boosting	0.768473	0.777962	0.768473	0.771760
3	Support Vector Classifier	0.724138	0.743129	0.724138	0.708454

XGBoost was selected as the final model because it achieved the best overall performance across the evaluation metrics, with 85.7% accuracy and an F1-score of 85.8%.

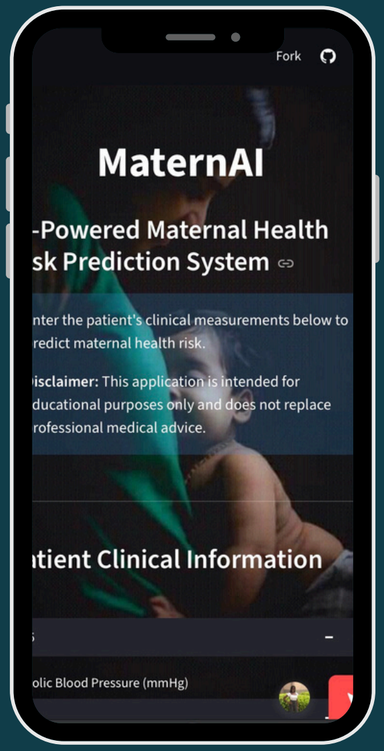
System Architecture

Link to App

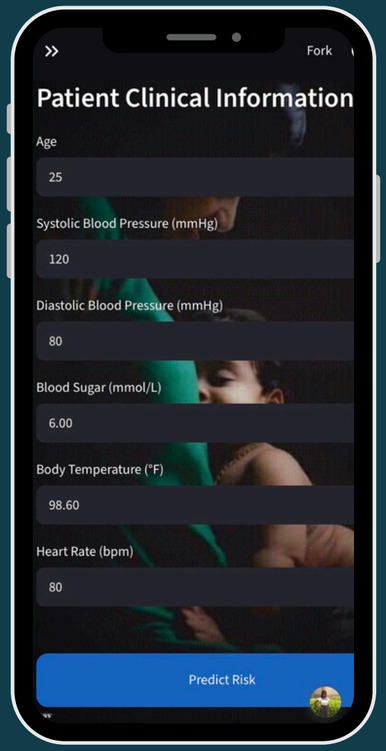
Here’s how the app would work:



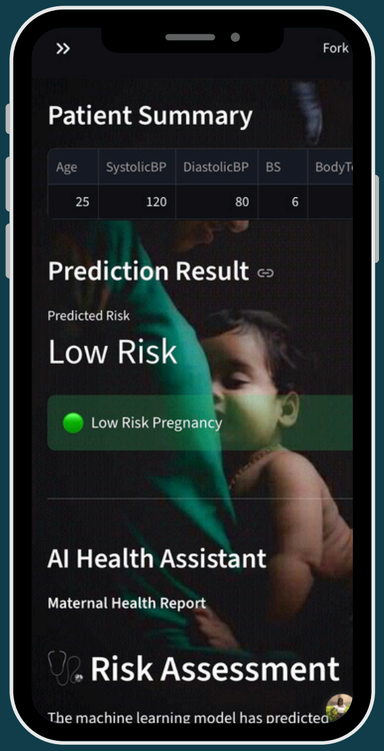
App Interface



Home Page



User Input



Prediction + AI Explanation

App Limitations

Not a Diagnostic Tool
Predicts risk category, not a medical diagnosis.

Limited Clinical Features
Uses six clinical measurements, excluding other potentially relevant factors.

Dataset Limitations
Performance depends on the quality and representativeness of the training data.

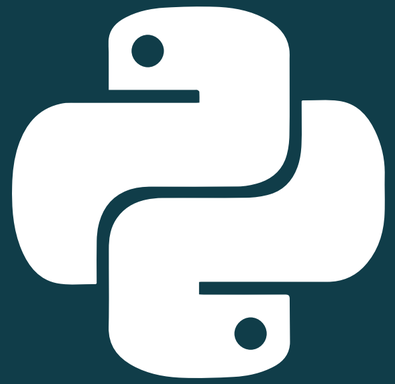
No Clinical Validation
Not yet tested in real-world clinical settings.



Tools Used



Google Colab – Data Cleaning, EDA, Model development and Evaluation



Python – For the source code for the App



Streamlit – Interactive Web Application and App Deployment

Thank You

Got questions ?



